

AI Store Generator

Multi-Tenant E-Commerce Platform

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Introduction

Project Definition

Souq Engine is an AI-powered SaaS platform for creating online stores.



Problem

Creating an online store quickly is challenging for small business owners and non-technical users.



Solution

The system uses AI and NLP to analyze the user's description and generate an initial store draft.



Output

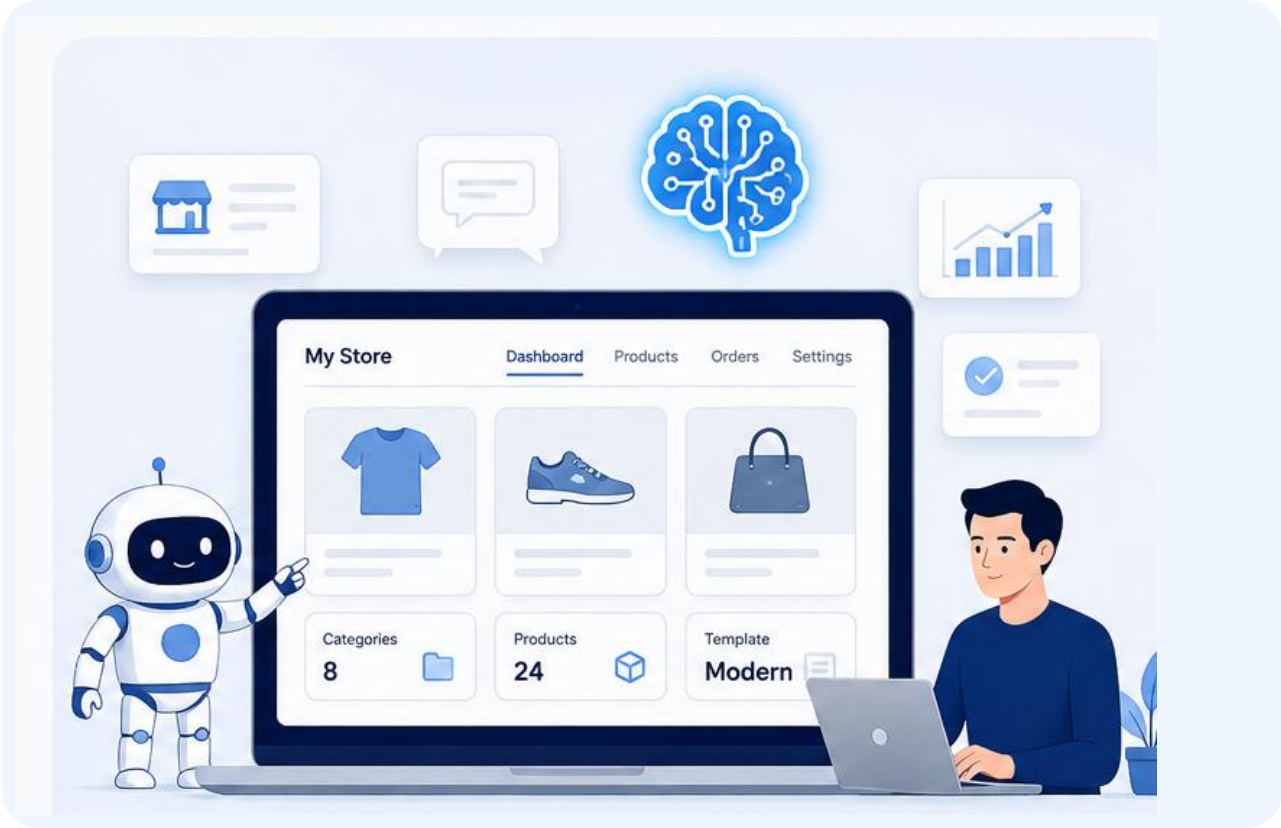
The draft includes store details, categories, mock products, and a suitable template with editing and validation options.



Key Result

A structured, editable store draft
draft generated within

3–5
seconds

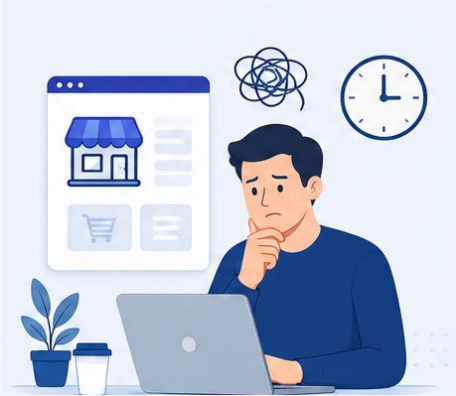


Problem Statement & Objectives



Problem

Creating an online store manually is time-consuming and difficult for small business owners and non-technical users.



Main Objective

Build an AI-assisted system that transforms a user's store description into a structured, editable store draft.



Generate Draft

Generate store details, categories, mock products, and a suitable template.



Clarify & Validate

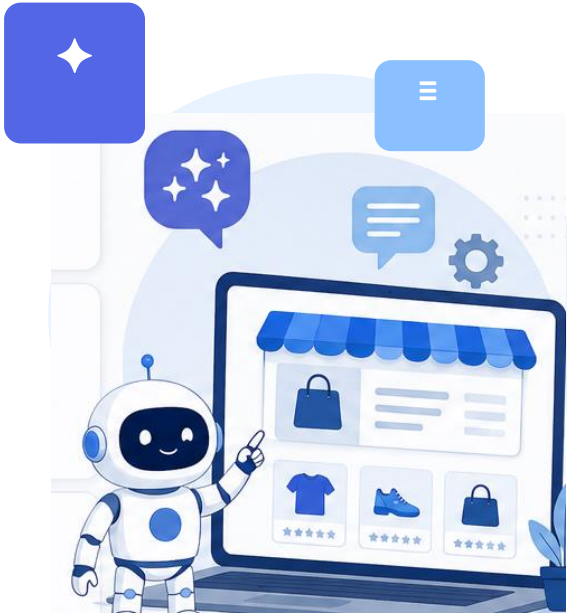
Ask clarification questions when the description is incomplete and validate the generated draft before final before final store creation.

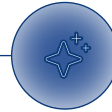


Performance

Generate the initial draft within

3–5 seconds.



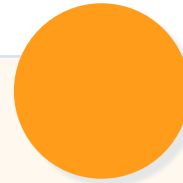


Scope, Constraints & Assumptions



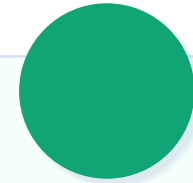
Scope

- Web-based SaaS platform supported by backend APIs. backend APIs.
- Multi-tenant architecture with auth and authorization. authorization.
- AI text-based store generator from a user description. description.
- Store, product, category, and theme management.
- Customer browsing with mock cart and basic checkout. checkout.
- Basic order management.
- Admin dashboard for management and monitoring. monitoring.
- Basic SEO and localization support.



Constraints

- Limited development time within one semester.
- Limited budget and development resources.
- Dependency on AI service availability and response time.
- No real payment or shipping integration in the MVP.
- Advanced AI image processing and chatbot automation are out of scope.



Assumptions

- Users have basic computer literacy.
- Internet connection is available.
- Store owners can provide a basic text description of their store idea.
- Users are registered and logged in to use store management features.
- AI-generated drafts are reviewed and edited before final creation.

Methodology and reference study



Related Work & Research Gap

Study / Platform	Contribution	Limitation
 Shopify	Provides a mature SaaS platform for building, customizing, and managing and managing online stores with integrations and themes.	Store setup still requires manual configuration and technical knowledge; it does not generate a complete store draft from a simple user description.
 Salla	Focuses on the Arabic market and supports Arabic language, local payment options, and easier store creation for regional users.	Limited AI-assisted capabilities for generating a full store draft with products and categories from the user description.
 Souq Engine (Our System)	Uses AI to generate an initial store draft from the user description, including products, categories, and suggested theme, with editing and validation.	Current MVP excludes real payment, shipping, advanced AI image processing, and chatbot automation.



Research Gap

Existing e-commerce platforms support store building and management, but they do not fully combine AI-based store generation, interactive clarification, editable draft validation, and multi-tenant SaaS management in one seamless workflow.



Our Innovation

Souq Engine transforms a simple text description into a structured, editable, and validated online store draft, while supporting store, product, category, order, and admin management in a multi-tenant SaaS platform tailored for the Arabic market.



Development Methodology & Project Plan



Methodology

Incremental Development

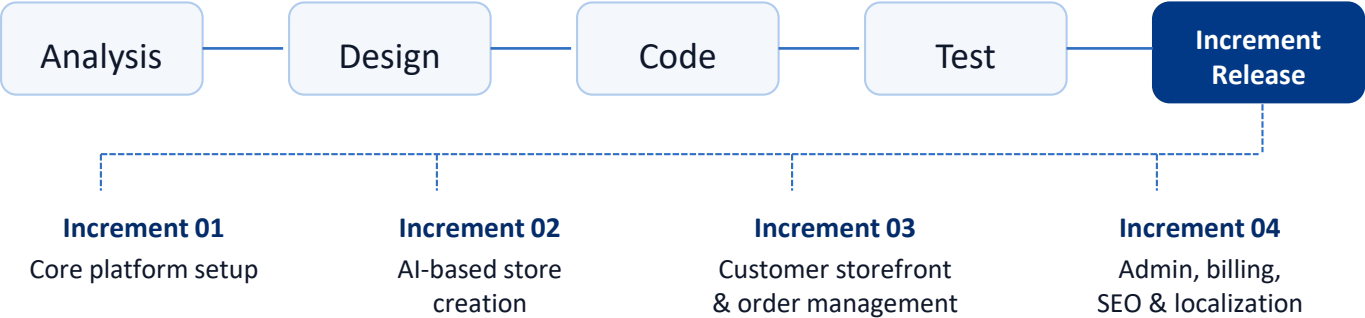
The project is developed in small, manageable increments. Each increment goes through analysis, design, implementation and testing before moving to the next.



Tools

- GitHub — Version Control
- GitHub Projects Task Tracking
- VS Code — Development
- Postman — API Testing
- Draw.io / Diagrams

Incremental Flow

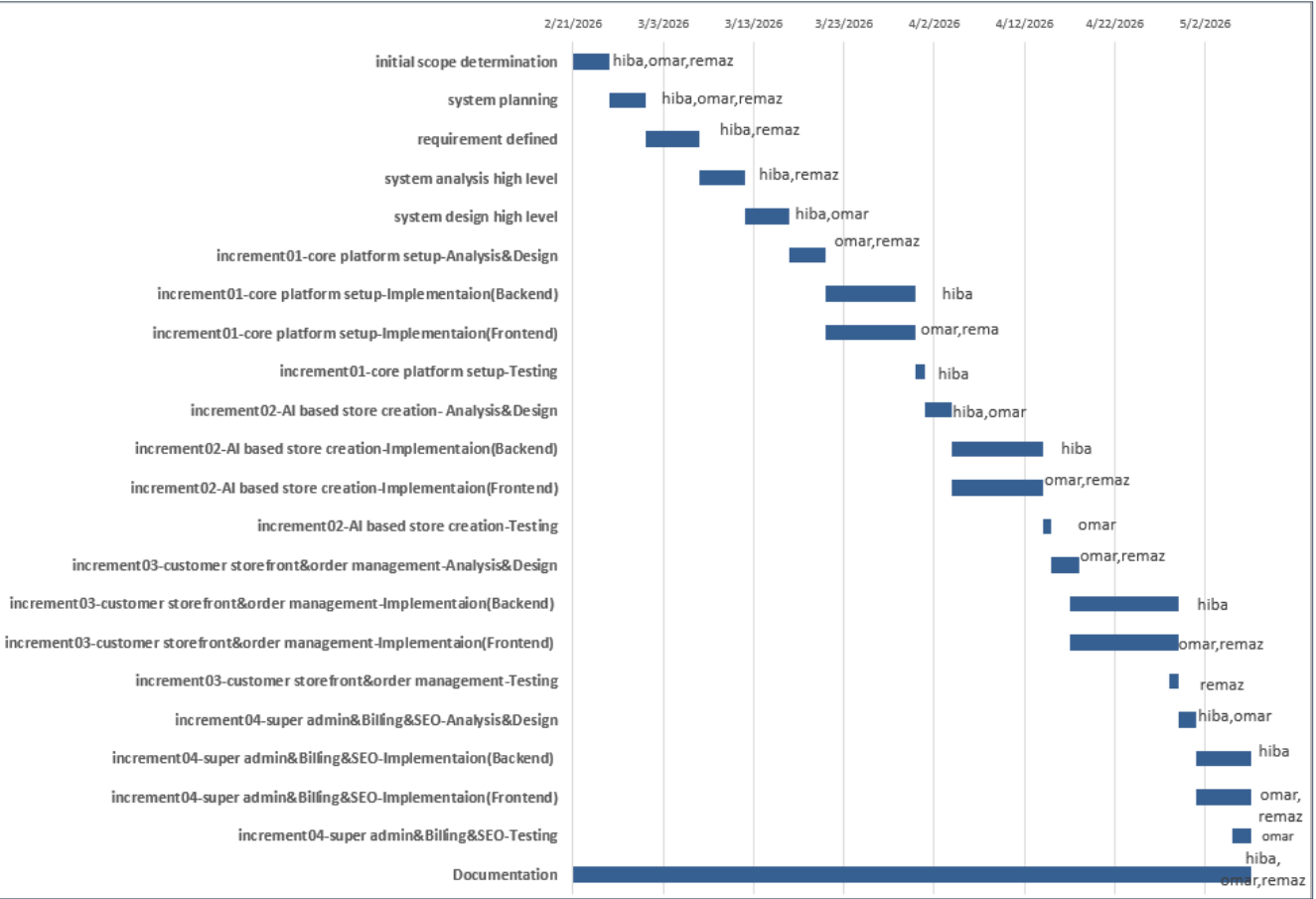


Project Timeline (High Level)

Week 1 - 2	Requirements Gathering & Scope Definition
Week 3 - 5	System Analysis & Design
Week 6 - 11	Implementation (Backend & Frontend)
Week 12 - 13	Testing & Evaluation
Week 14 - 15	Documentation

Gantt Chart

Task Name	Duration(days)	Start Date	End Date
initial scope determination	4	2/21/2026	2/24/2026
system planning	4	2/25/2026	2/28/2026
requirement defined	6	3/1/2026	3/6/2026
system analysis high level	5	3/7/2026	3/11/2026
system design high level	5	3/12/2026	3/16/2026
increment01-core platform setup-Analysis&Design	4	3/17/2026	3/20/2026
increment01-core platform setup-Implementaion(Backend)	10	3/21/2026	3/30/2026
increment01-core platform setup-Implementaion(Frontend)	10	3/21/2026	3/30/2026
increment01-core platform setup-Testing	1	3/31/2026	3/31/2026
increment02-AI based store creation- Analysis&Design	3	4/1/2026	4/3/2026
increment02-AI based store creation-Implementaion(Backend)	10	4/4/2026	4/13/2026
increment02-AI based store creation-Implementaion(Frontend)	10	4/4/2026	4/13/2026
increment02-AI based store creation-Testing	1	4/14/2026	4/14/2026
increment03-customer storefront&order management-Analysis&Design	3	4/15/2026	4/17/2026
increment03-customer storefront&order management-Implementaion(Backend)	12	4/17/2026	4/28/2026
increment03-customer storefront&order management-Implementaion(Frontend)	12	4/17/2026	4/28/2026
increment03-customer storefront&order management-Testing	1	4/28/2026	4/28/2026
increment04-super admin&Billing&SEO-Analysis&Design	2	4/29/2026	4/30/2026
increment04-super admin&Billing&SEO-Implementaion(Backend)	7	5/1/2026	5/7/2026
increment04-super admin&Billing&SEO-Implementaion(Frontend)	7	5/1/2026	5/7/2026
increment04-super admin&Billing&SEO-Testing	4	5/5/2026	5/8/2026
Documentation	77	2/21/2026	5/8/2026



System Analysis



Functional Requirements

The system provides the following core functions to support store owners, customers, and administrators.



Store Owner

- 01** Register, login, and manage account information.
- 02** Create a store using AI from a text description.
- 03** Review, edit, regenerate, and confirm AI-generated store draft.
- 04** Manage store settings, themes, domains, and publishing status.
- 05** Manage products, categories, images, prices, and inventory.
- 06** View and manage customer orders and update order status.



Customer

- 01** Browse the public store.
- 02** View categories and product details.
- 03** Add products to the mock cart.
- 04** Create a basic order from the mock mock cart.



Super Admin

- 01** Login to the admin dashboard.
- 02** Manage platform stores and users.
- 03** Monitor platform data and update global settings.



Non-Functional Requirements

The system is designed to meet the following quality attributes and operational requirements.

P	Performance	Standard APIs should respond quickly, while AI store draft generation targets 3–5 seconds.
SEC	Security	JWT authentication, authorization, input validation, encryption, and tenant-based access control are applied.
DI	Data Isolation	Each store owner can access only their own store data using tenant/store scoping.
U	Usability	The interface should be simple, intuitive, and suitable for non-technical users.
A	Availability	The system provides fallback options when the AI service fails or times out, and aims for high availability.
S	Scalability	Layered backend architecture supports future horizontal scaling and increased load.
M	Maintainability	The system separates AI logic, validation, business logic, and APIs for easier updates and maintenance.



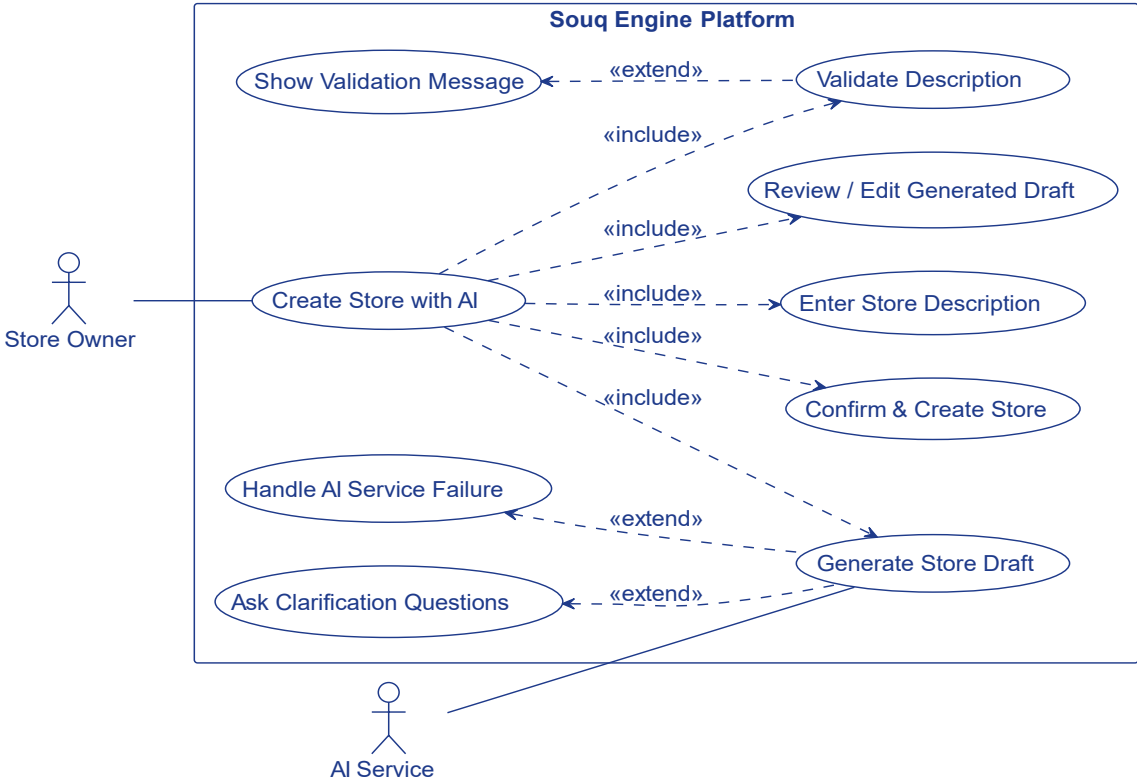


UML Diagrams — Use Case Example

U Use Case Summary

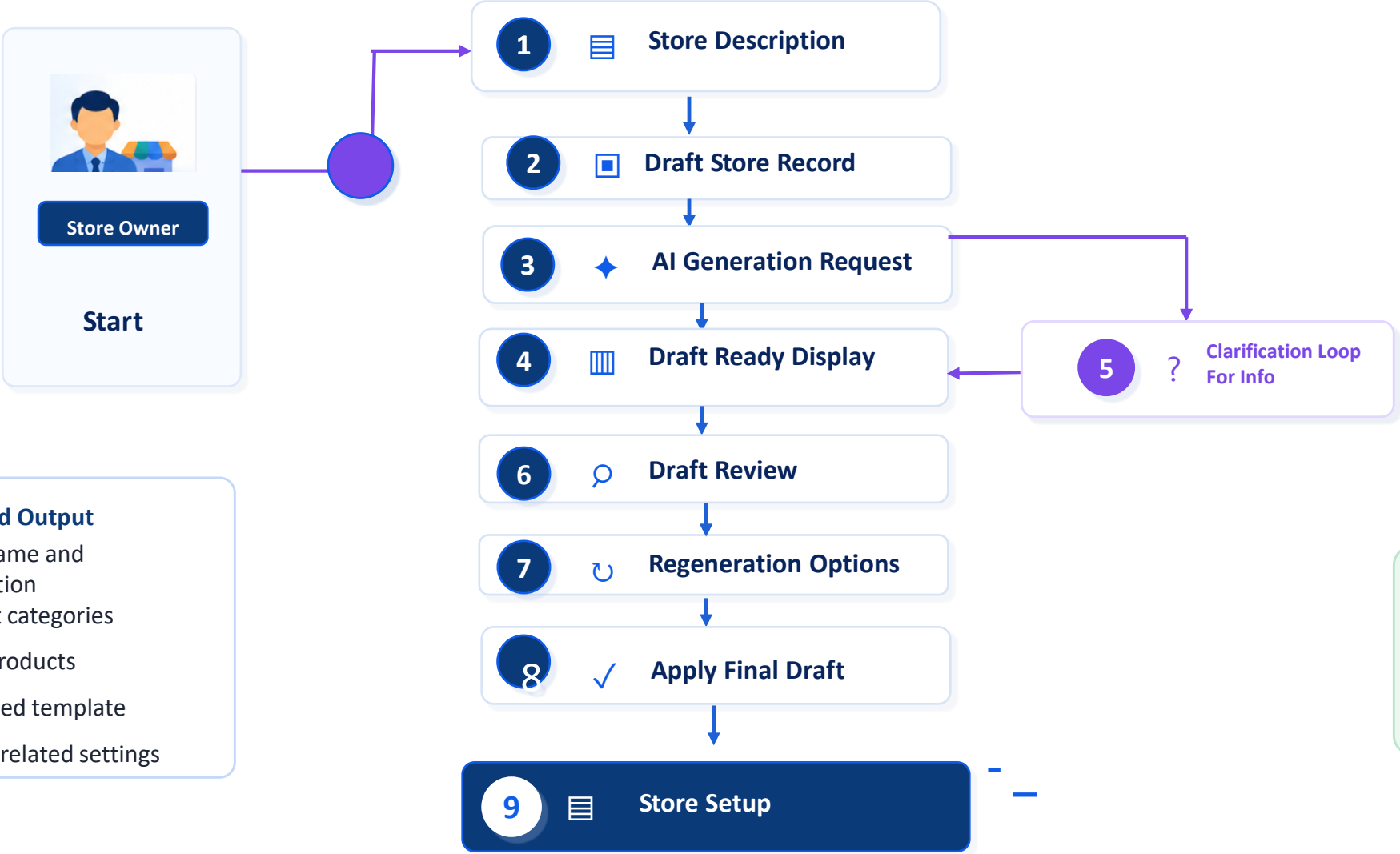
Use Case	AI-Assisted Store Creation
Actor	Store Owner
Description	The store owner enters a store description, and the system uses AI to generate basic store settings.
Precondition	The store owner is registered and logged in.
Main Flow	Enter description → Validate input → Generate AI draft → Review/Edit → Confirm store creation
Alternative Flows	Short description, incomplete input, AI service failure Description Insufficient →MCQ questions
Postcondition	A new draft store is created and linked to the user account.

Use Case Diagram: AI Store Creation



Intelligent Algorithm Implementation

AI Store Creation Pipeline



- Generated Output**
- Store name and description
 - Product categories
 - Mock products
 - Suggested template
 - Theme-related settings

Approach

Single Call with AI-Driven Conditional Clarification and Interactive Refinement.

Key Features

- AI assesses if the description is sufficient. Clarification questions if information is missing.
- Interactive review, manual edit, and regeneration.
- Validation ensures data quality and consistency.

Result

Structured, editable store store draft generated within **3–5 seconds.**

System Design

 Presentation Layer

React-based web interface for store owners, customers, and admins.

 Business Layer

Manages authentication, tenant isolation, store operations, products, categories, orders, and validation rules.

 Data Layer

PostgreSQL

System Architecture

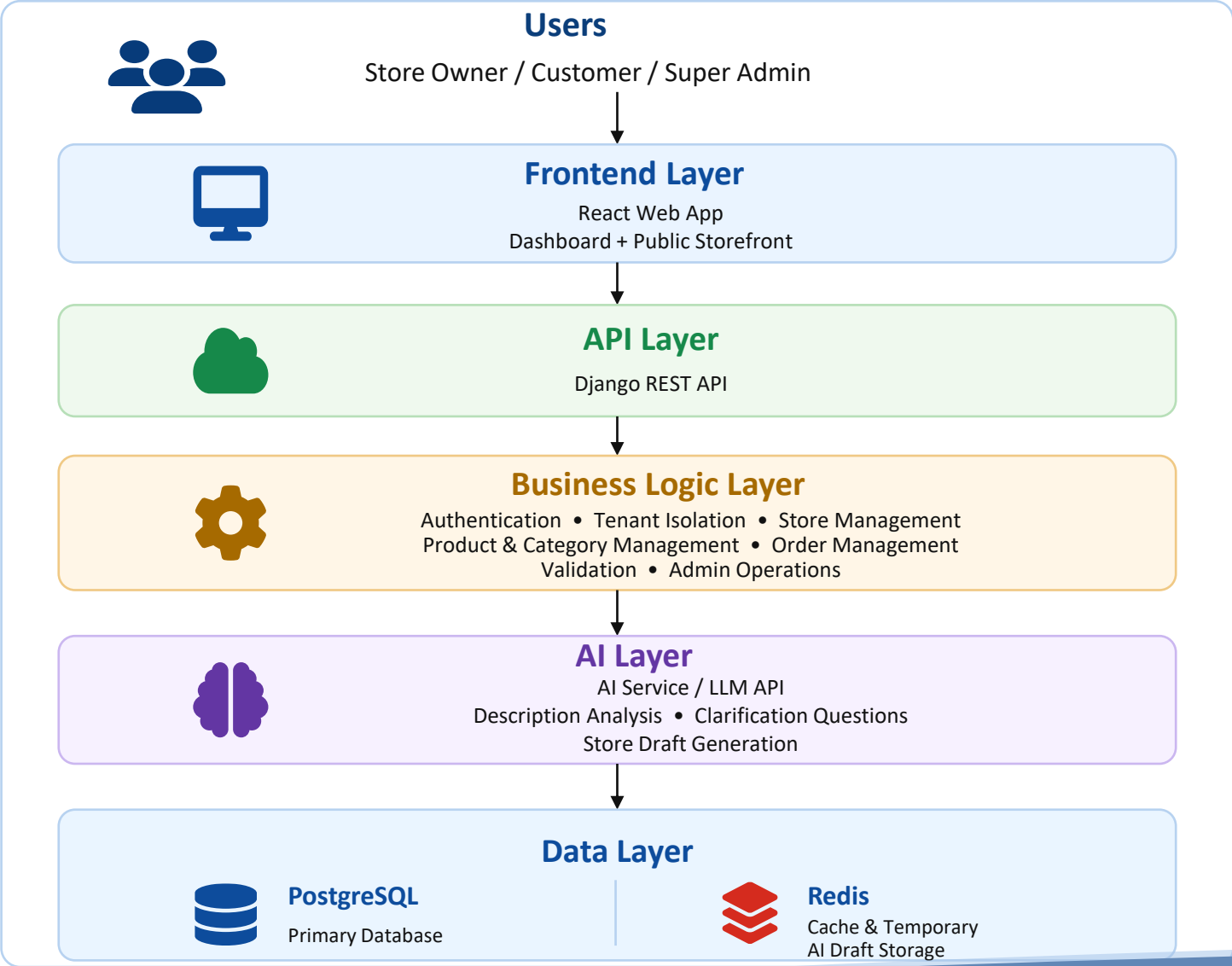
Layered Architecture
Architecture with AI
AI Service Integration
Integration

 API Layer

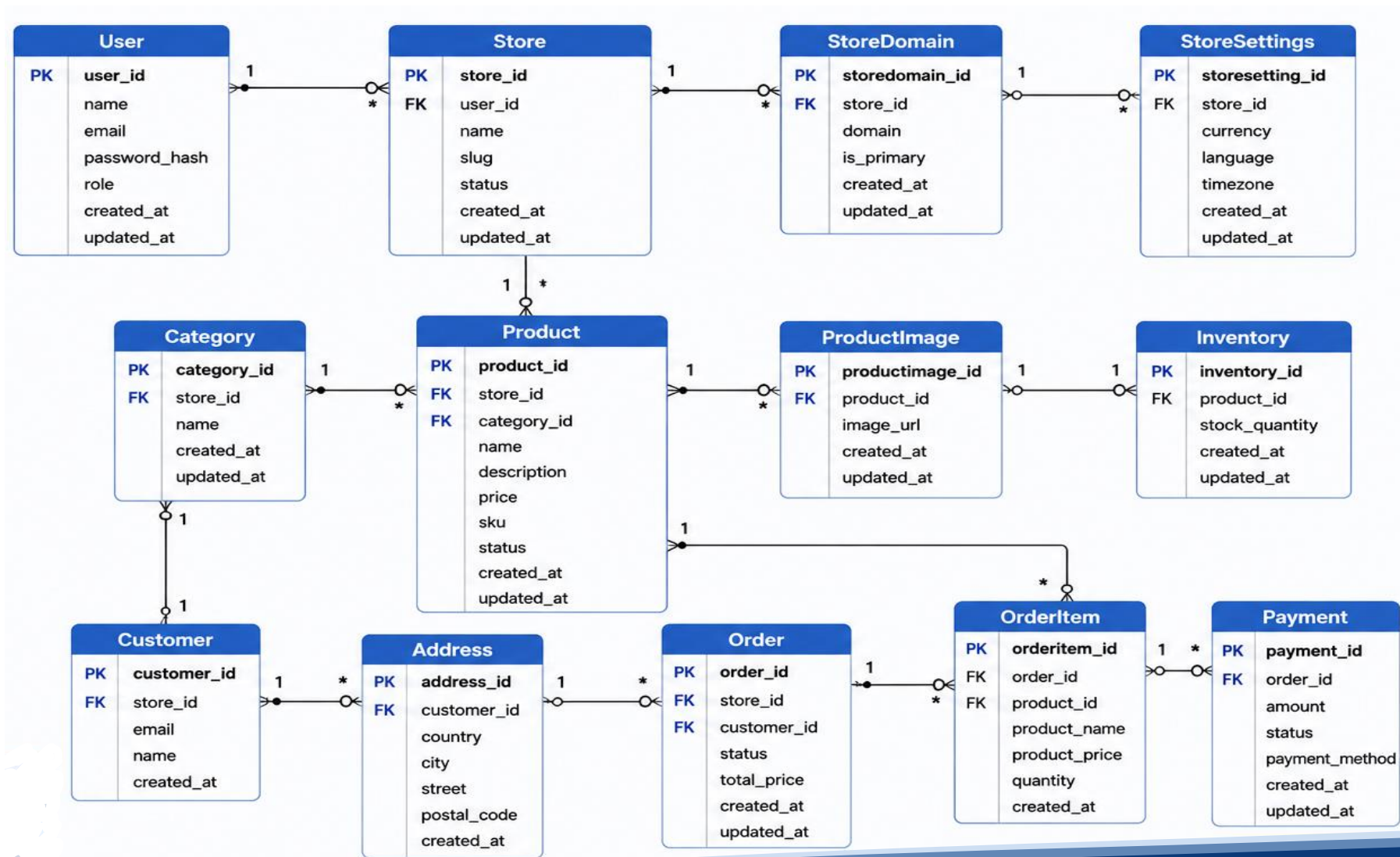
Django REST API handles DRF, simple JWT , tenant middleware, APIs docs

 AI Layer

AI service analyzes the user’s store description, asks clarification questions when needed, and generates the initial store draft.



Databa/Data Model



Implementation



User Interface

Sign in to manage your stores and connect the real backend workspace

After sign in, the frontend will bootstrap your user session and load your available stores.

العربية

Welcome Back

Sign in to your account to continue

Email

✉

usertest1@gmail.com

Password

🔒

.....

👁

☐ Remember me

Login →

Don't have an account? Register

Q Search

Ctrl+K

My Test Store my-test-store-1

Dashboard

AI Store Generator

Products

Categories

Orders

Customers

Appearance

Store Settings

Billing

Shop

Dashboard

Welcome back, usertest1

Last 7 Days

AI Store Generator

Total Orders

0

+12.5% vs last month

Total Revenue

\$0

+8.2% vs last month

Total Products

1

+3.1% vs last month

Total Customers

0

-2.4% vs last month

Recent Orders

View All →

Top Products

View All →



User Interface

SOUQ ENGINE

My Test Store

Dashboard

AI Store Generator

Products

Categories

Orders

Customers

Appearance

Store Settings

Billing

Shop

Search

EN

usertest1

AI store draft assistant

Start from an idea, then review the draft before approval

AI creates a reviewable store draft only. Nothing is treated as final or persistence-ready until you review, edit, and confirm it.

Describe the store idea

أريد انشاء متجر للملابس الرجالية من عمر 15 حتى 50 و اريد ان تصنع لي المتجر باللونين الأبيض و الأزرق

99/500

Generate store draft

Example prompts

Modern Arabic skincare store for sensitive skin with a calm premium look and gift-ready sets

Streetwear store for university students with bold drops, limited collections, and fast local delivery

Home bakery storefront for custom cakes, celebration boxes, and seasonal sweets across Damascus

Shop

SOUQ ENGINE

متجر الأناقة الرجالية

Dashboard

AI Store Generator

Products

Categories

Orders

Customers

Appearance

Store Settings

Billing

Shop

Search

EN

usertest1

Products

Manage your product catalog, SKU, stock, images, and backend-backed details.

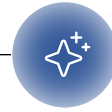
Search

Filter

All Products (3)

Product	Category	Price	Stock	Status	Actions
سترة جلدية بنية SKU: MC-JCK-001 سترة جلدية رجالية بنية، تصميم عصري، سحاب أمامي.	سترة	\$120.00	15	Active	...
بنطلون جينز أسود SKU: MC-JE-001 بنطلون جينز رجالي بقصة مستقيمة، لون أسود، خياطة عالية الجودة.	بنطلون	\$45.00	30	Active	...
قميص كاجوال أزرق SKU: MC-TSH-001 قميص كاجوال بأكمام طويلة، قماش قطني ناعم، لون أزرق داكن.	قميص	\$35.00	50	Active	...

Shop



Implementation & Testing Tools

Key technologies used to build, test, and manage Souq Engine



Frontend

React

Used for building the user interface and dashboards.



Backend

django

Django (Python) for REST APIs and business logic.



Database

PostgreSQL

Relational database for persistent data storage.



Cache / Session

redis

Redis for caching, sessions, and AI draft storage.



API Testing

POSTMAN

Used for testing and validating API endpoints and responses.



Automated Testing

pytest

Unit, API, integration, and regression testing.



Version Control

git GitHub

Source code management with Git and GitHub.



Other Tools

Python JWT DRF

Libraries and tools that support development and security.

Testing



Testing & Evaluation Results

TEST EXECUTION SUMMARY	
Total Tests	396
Passed Tests	396
Failed Tests	0
Skipped Tests	1
Pass Rate (for executed tests)	100%
Execution Time	1281.180 sec (≈ 20 min)
Final Status	OK

QUALITY METRICS		
</>	Automated Backend Tests	396
🛡️	Overall Code Coverage	78%
✓	CI / Test Status	Passed



Req_ID	Use cases	Title	analysis	System design	Detailed design	coding	test cases
FR-01	UC-AI-01	Enter Store Description	AI-Assisted Store Creation-analysis	System Design	Detail design	Source code-backend Source code frontend	AI-Assisted Store Creation-TC-01
FR-02	UC-AI-01	Send Description to AI Service	AI-Assisted Store Creation-analysis	System Design	Detail design	Source code-backend Source code frontend	AI-Assisted Store Creation-TC-02
FR-03	UC-SM-01	Create Store	Store Management-analysis	System Design	Detail design	Source code-backend Source code frontend	Store Management-TC-05



Comparative Analysis & Objectives Met

Souq Engine simplifies store creation using AI and meets all key project objectives.

1. Comparative Analysis

Criteria	Traditional Store Builders (Shopify / Salla / Others)	Souq Engine (Our System)
 Store Creation	Mostly manual setup and configuration	AI-assisted store draft generation
 Input Method	Forms and manual configuration	Simple text description
 Clarification Questions	Not provided	AI asks clarification questions when information is incomplete
 Generated Content	User adds data manually	Store details, categories, mock products, and template
 Editing	Supported	Supported
 Arabic / Arab Market Support	Supported in some systems	Fully considered in system scope

2. Objectives Met

 AI-assisted store creation	Achieved 	Ensure tenant data isolation and access control	Achieved
 Generate editable store draft from user description			
 Support clarification questions for incomplete input	Achieved 	Automated backend testing	Achieved — 396 passed tests
 Validate generated draft before store creation	Achieved 	Code coverage	Achieved — 78%

Conclusion



Achievements

- Built Souq Engine, an AI-assisted SaaS platform for creating online stores
- Generated editable store drafts from simple user descriptions
- Supported store, product, category, order, theme, and admin management
- Achieved 396 passed backend tests with 78% code coverage



Challenges

- Handling incomplete or vague user descriptions
- Validating AI-generated store drafts before saving
- Ensuring tenant-based data isolation and secure access
- Limited development time within one semester



Future Work

- Add real payment gateway integration
- Improve AI-generated product images and descriptions
- Add advanced analytics dashboard
- Deploy the platform on cloud infrastructure



Final Recommendation

AI-assisted store creation can reduce setup effort and help small businesses launch faster.



*Thank you
for listening*

